

# In-frame deletion of lasB in LB001 via pEXG2 allelic exchange

Primer design • PCR / Assembly protocol • 2026-05-06

## 1 Konstrukt-Überblick

|                  |                                            |
|------------------|--------------------------------------------|
| Zielgen          | lasB in LB001 (1497 nt, 498 aa)            |
| Strategie        | In-frame Deletion, Scar 18 aa (16 N + 2 C) |
| Vektor           | pEXG2 (5084 bp), HindIII-linearisiert      |
| Methode          | In-Fusion 3-Fragment (Vektor + UP + DN)    |
| Counterselektion | sacB / GmR (pEXG2)                         |
| Finales Plasmid  | pEXG2-ΔlasB_LB001 • 6155 bp                |

## 2 Primer

| Primer         | Sequenz 5'→ 3' (Tail fett, Body normal) | nt | Tm <sub>body</sub> | GC   |
|----------------|-----------------------------------------|----|--------------------|------|
| P1_lasB_UP_Fwd | CATAAATGTAAAGCAGGTGTTCCAGCTGGTGCGAG     | 34 | 60.9 °C            | 63 % |
| P2_lasB_UP_Rev | CCGAGCTTACAACGCACCCATGATCGCAACGAACAAC   | 37 | 60.8 °C            | 50 % |
| P3_lasB_DN_Fwd | GTTGCGATCATGGGTGCGTTGTAAGCTCGGTGGTC     | 35 | 61.0 °C            | 60 % |
| P4_lasB_DN_Rev | CGACCTGCAGAAGCTGCCAGGTACTCGCCTTGC       | 33 | 60.7 °C            | 67 % |

## 3 PCR-Bedingungen (B7 High Fidelity)

|                           |                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------|
| Tm <sub>body</sub> Mittel | 60.9 °C (Spread 0.21 °C)                                                              |
| Annealing Ta              | 61 °C (= Tm <sub>body</sub> ; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt) |
| Elongation                | 60 s/kb • UP 593 bp → 36 s • DN 538 bp → 32 s                                         |

## 4 Konstrukt-Zusammenfassung

UP-Amplicon 593 bp • DN-Amplicon 538 bp • Insert 1101 bp • Final 6155 bp

### Scar-Sequenz (16 N + 2 C aa + Stop):

ATG AAG AAG GTT TCT ACG CTT GAC CTG TTG TTC GTT GCG ATC ATG GGT | GCG TTG TAA  
M K K V S T L D L L F V A I M G | A L \*

Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 0 (erwartet 0)